Supplementary Material

KAGNet: Knowledge-Enriched Multi-PPI Learning for Cancer Driver Gene Prioritization

Table S1. Summary of input feature dimensionality and gene counts across the 12 cancer types.

Category	BRCA	BLCA	LUAD	THCA	LIHC	LUSC	ESCA	PRAD	STAD	COAD	UCEC	CESC
Multi-omics features	4	4	4	4	4	4	4	4	4	4	4	4
Driver genes	202	95	179	64	82	26	71	50	73	141	68	17
Passenger genes	2187	2187	2187	2187	2187	2187	2187	2187	2187	2187	2187	2187

Table S2. AUPRC comparison across 12 cancer types with EMGNN under its original input preprocessing. All columns except EMGNN are identical to the AUPRC benchmark in the main text; here, EMGNN is trained on multi-omics features without z-score normalization. Results are reported as mean AUPRC $\pm$ standard deviation over 10 runs of stratified 5-fold cross-validation. MRNGCN results are taken from the original publication. The best result for each cancer type is shown in **bold**.

Cancer	KAGNet	EMOGI	MTGCN	DGHNN	MRNGCN	MODIG	EMGNN
BRCA	**0.835 ± 0.004**	0.648 ± 0.005	0.658 ± 0.004	0.683 ± 0.006	0.692 ± 0.004	0.586 ± 0.005	0.676 ± 0.010
BLCA	**0.798 ± 0.008**	0.549 ± 0.010	0.657 ± 0.008	0.694 ± 0.006	0.725 ± 0.005	0.681 ± 0.025	0.718 ± 0.018
LUAD	**0.742 ± 0.006**	0.559 ± 0.011	0.628 ± 0.010	0.641 ± 0.009	0.629 ± 0.012	0.603 ± 0.003	0.633 ± 0.019
LIHC	**0.619 ± 0.017**	0.385 ± 0.019	0.465 ± 0.020	0.481 ± 0.019	0.547 ± 0.015	0.253 ± 0.025	0.389 ± 0.024
THCA	**0.573 ± 0.017**	0.259 ± 0.008	0.291 ± 0.011	0.315 ± 0.023	0.308 ± 0.010	0.028 ± 0.001	0.208 ± 0.037
LUSC	**0.705 ± 0.047**	0.222 ± 0.027	0.310 ± 0.041	0.405 ± 0.054	0.348 ± 0.032	0.071 ± 0.014	0.228 ± 0.023
ESCA	**0.639 ± 0.012**	0.417 ± 0.030	0.472 ± 0.028	0.476 ± 0.015	0.537 ± 0.012	0.264 ± 0.019	0.347 ± 0.035
PRAD	**0.852 ± 0.008**	0.523 ± 0.022	0.614 ± 0.019	0.706 ± 0.013	0.669 ± 0.016	0.383 ± 0.097	0.555 ± 0.032
STAD	**0.755 ± 0.011**	0.460 ± 0.015	0.593 ± 0.012	0.645 ± 0.008	0.646 ± 0.011	0.552 ± 0.051	0.550 ± 0.029
COAD	**0.630 ± 0.009**	0.346 ± 0.007	0.382 ± 0.009	0.447 ± 0.069	0.495 ± 0.007	0.443 ± 0.066	0.424 ± 0.031
UCEC	**0.761 ± 0.010**	0.400 ± 0.014	0.495 ± 0.015	0.534 ± 0.025	0.570 ± 0.015	0.389 ± 0.080	0.452 ± 0.014
CESC	**0.717 ± 0.036**	0.555 ± 0.052	0.597 ± 0.053	0.564 ± 0.041	0.534 ± 0.069	0.059 ± 0.046	0.416 ± 0.033

Table S3. Comparison of Hits@K across 12 cancer types for $K = 1, 3, 5, 10, 20, 50$ with EMGNN under its original input preprocessing. All rows except EMGNN are identical to the full Hits@K comparison in this Supplementary Material; here, EMGNN is trained on multi-omics features without z-score normalization. Higher values indicate greater recovery of known driver genes among the top-K ranked candidates. MRNGCN is not included because its publicly available implementation could not be executed. The best result for each cancer type at each K is shown in **bold**.

K	Method	BRCA	BLCA	LUAD	LIHC	THCA	LUSC	ESCA	PRAD	STAD	COAD	UCEC	CESC
1	KAGNet	**0.66**	0.28	0.42	**0.28**	**0.48**	**0.18**	**0.78**	**0.40**	**0.48**	**0.92**	**0.78**	**0.24**
	EMOGI	0.26	0.16	0.06	0.08	0.00	0.10	0.04	0.12	0.06	0.16	0.06	0.00
	MTGCN	0.20	0.16	0.20	0.08	0.04	0.02	0.02	0.04	0.04	0.02	0.06	0.00
	DGHNN	0.20	0.10	0.20	0.04	0.02	0.04	0.20	0.24	0.22	0.16	0.42	0.06
	MODIG	0.32	0.30	0.28	0.12	0.00	0.00	0.40	0.12	0.14	0.40	0.38	0.00
	EMGNN	0.06	**0.40**	**0.54**	0.08	0.00	0.00	0.00	0.06	0.00	0.02	0.00	0.00
3	KAGNet	**1.48**	**1.66**	0.70	**0.86**	**0.76**	0.20	**1.42**	**1.22**	**1.16**	**2.16**	**1.34**	**0.46**
	EMOGI	0.60	0.34	0.34	0.20	0.00	0.10	0.14	0.22	0.30	0.34	0.38	0.00
	MTGCN	0.38	0.24	0.26	0.16	0.08	0.02	0.18	0.08	0.22	0.18	0.12	0.06
	DGHNN	0.62	0.52	0.32	0.24	0.08	**0.40**	0.32	0.42	0.38	0.22	0.48	0.08
	MODIG	0.92	1.00	0.56	0.20	0.00	0.00	0.46	0.26	0.40	1.12	0.60	0.00
	EMGNN	0.50	1.56	**0.86**	0.34	0.18	0.12	0.10	0.42	0.28	0.14	0.26	0.00
5	KAGNet	**2.30**	**2.62**	1.06	**1.00**	**1.02**	0.36	**1.70**	**1.54**	**1.36**	**2.60**	**1.56**	**0.64**
	EMOGI	0.96	0.42	0.38	0.30	0.00	0.12	0.14	0.32	0.64	0.42	0.48	0.00
	MTGCN	0.80	0.36	0.50	0.22	0.08	0.04	0.26	0.18	0.50	0.40	0.26	0.14
	DGHNN	1.06	0.84	0.42	0.28	0.22	**0.64**	0.44	0.86	0.58	0.36	0.82	0.12
	MODIG	0.92	1.20	0.68	0.26	0.00	0.00	0.54	0.52	0.72	1.46	0.64	0.00
	EMGNN	1.00	2.44	**1.08**	0.38	0.18	0.16	0.34	0.54	0.48	0.18	0.34	0.14
10	KAGNet	**4.14**	**4.26**	**1.62**	**1.30**	**1.10**	0.58	**2.04**	**1.98**	**1.58**	**3.38**	**2.30**	**0.80**
	EMOGI	1.44	0.56	0.78	0.36	0.20	0.14	0.32	0.44	0.76	0.56	0.74	0.02
	MTGCN	1.66	0.56	0.84	0.24	0.16	0.08	0.30	0.36	0.64	0.58	0.50	0.24
	DGHNN	1.64	1.42	0.62	0.34	0.26	**0.64**	0.62	1.82	1.02	0.48	1.48	0.44
	MODIG	1.52	1.80	0.96	0.46	0.00	0.00	0.72	0.92	1.26	1.66	0.88	0.00
	EMGNN	1.78	3.62	1.52	0.48	0.20	0.22	0.46	0.80	1.08	0.38	0.86	0.20
20	KAGNet	**5.42**	**5.68**	2.06	**2.02**	**1.28**	**0.76**	**2.60**	**2.84**	**3.04**	**4.28**	**3.32**	**1.08**
	EMOGI	1.82	0.62	1.04	0.46	0.26	0.16	0.40	0.52	0.86	0.80	1.04	0.04
	MTGCN	2.20	0.72	1.34	0.24	0.22	0.12	0.56	0.82	0.76	0.72	0.76	0.34
	DGHNN	2.70	2.14	1.52	0.62	0.26	0.66	0.86	2.66	1.84	0.62	1.86	0.44
	MODIG	2.40	3.40	1.46	0.60	0.00	0.00	0.86	1.00	1.86	2.12	1.02	0.08
	EMGNN	2.40	4.70	**2.28**	0.72	0.34	0.26	0.78	1.32	1.68	1.00	1.28	0.22
50	KAGNet	**8.76**	**7.32**	**4.30**	**3.44**	**1.78**	**0.96**	**3.30**	**4.22**	**4.12**	**5.18**	**4.74**	**1.36**
	EMOGI	3.06	1.34	2.04	0.76	0.44	0.20	0.92	1.08	1.36	1.54	1.40	0.34
	MTGCN	3.20	1.54	2.22	0.40	0.26	0.22	0.96	1.26	1.46	1.14	1.08	0.48
	DGHNN	4.52	3.28	3.64	1.46	0.48	0.72	1.12	3.82	3.28	0.94	2.86	0.84
	MODIG	3.96	5.80	3.46	1.00	0.00	0.08	1.46	1.72	3.26	3.32	1.70	0.12
	EMGNN	3.90	6.32	3.36	1.20	0.34	0.32	1.40	1.92	2.70	1.72	1.70	0.40

Table S4. Comparison of Hits@K across 12 cancer types for $K = 1, 3, 5, 10, 20, 50$. EMGNN is trained on z-score normalized multi-omics features, as in the main text. Higher values indicate greater recovery of known driver genes among the top-K ranked candidates. MRNGCN is not included because its publicly available implementation could not be executed. The best result for each cancer type at each K is shown in **bold**.

K	Method	BRCA	BLCA	LUAD	LIHC	THCA	LUSC	ESCA	PRAD	STAD	COAD	UCEC	CESC
1	KAGNet	**0.66**	0.28	**0.42**	**0.28**	**0.48**	**0.18**	**0.78**	**0.40**	**0.48**	**0.92**	**0.78**	**0.24**
	EMOGI	0.26	0.16	0.06	0.08	0.00	0.10	0.04	0.12	0.06	0.16	0.06	0.00
	MTGCN	0.20	0.16	0.20	0.08	0.04	0.02	0.02	0.04	0.04	0.02	0.06	0.00
	DGHNN	0.20	0.10	0.20	0.04	0.02	0.04	0.20	0.24	0.22	0.16	0.42	0.06
	MODIG	0.32	**0.30**	0.28	0.12	0.00	0.00	0.40	0.12	0.14	0.40	0.38	0.00
	EMGNN	0.40	0.26	0.34	0.20	0.36	**0.18**	0.58	0.26	0.34	0.86	0.52	0.18
3	KAGNet	**1.48**	**1.66**	**0.70**	**0.86**	**0.76**	0.20	**1.42**	**1.22**	**1.16**	**2.16**	**1.34**	**0.46**
	EMOGI	0.60	0.34	0.34	0.20	0.00	0.10	0.14	0.22	0.30	0.34	0.38	0.00
	MTGCN	0.38	0.24	0.26	0.16	0.08	0.02	0.18	0.08	0.22	0.18	0.12	0.06
	DGHNN	0.62	0.52	0.32	0.24	0.08	**0.40**	0.32	0.42	0.38	0.22	0.48	0.08
	MODIG	0.92	1.00	0.56	0.20	0.00	0.00	0.46	0.26	0.40	1.12	0.60	0.00
	EMGNN	1.08	1.40	0.66	0.78	0.66	0.30	1.18	0.96	1.14	1.80	1.06	0.20
5	KAGNet	**2.30**	**2.62**	**1.06**	**1.00**	**1.02**	0.36	**1.70**	**1.54**	**1.36**	**2.60**	1.56	0.64
	EMOGI	0.96	0.42	0.38	0.30	0.00	0.12	0.14	0.32	0.64	0.42	0.48	0.00
	MTGCN	0.80	0.36	0.50	0.22	0.08	0.04	0.26	0.18	0.50	0.40	0.26	0.14
	DGHNN	1.06	0.84	0.42	0.28	0.22	**0.64**	0.44	0.86	0.58	0.36	0.82	0.12
	MODIG	0.92	1.20	0.68	0.26	0.00	0.00	0.54	0.52	0.72	1.46	0.64	0.00
	EMGNN	1.92	2.28	0.96	0.92	0.68	0.36	1.30	1.10	**1.36**	2.14	**1.60**	**0.82**
10	KAGNet	**4.14**	**4.26**	**1.62**	**1.30**	**1.10**	0.58	**2.04**	**1.98**	1.58	**3.38**	2.30	0.80
	EMOGI	1.44	0.56	0.78	0.36	0.20	0.14	0.32	0.44	0.76	0.56	0.74	0.02
	MTGCN	1.66	0.56	0.84	0.24	0.16	0.08	0.30	0.36	0.64	0.58	0.50	0.24
	DGHNN	1.64	1.42	0.62	0.34	0.26	**0.64**	0.62	1.82	1.02	0.48	1.48	0.44
	MODIG	1.52	1.80	0.96	0.46	0.00	0.00	0.72	0.92	1.26	1.66	0.88	0.00
	EMGNN	3.44	3.98	1.42	1.28	0.84	0.50	1.78	1.48	**1.90**	2.82	**2.40**	**1.06**
20	KAGNet	**5.42**	**5.68**	**2.06**	**2.02**	**1.28**	**0.76**	2.60	**2.84**	**3.04**	**4.28**	**3.32**	1.08
	EMOGI	1.82	0.62	1.04	0.46	0.26	0.16	0.40	0.52	0.86	0.80	1.04	0.04
	MTGCN	2.20	0.72	1.34	0.24	0.22	0.12	0.56	0.82	0.76	0.72	0.76	0.34
	DGHNN	2.70	2.14	1.52	0.62	0.26	0.66	0.86	2.66	1.84	0.62	1.86	0.44
	MODIG	2.40	3.40	1.46	0.60	0.00	0.00	0.86	1.00	1.86	2.12	1.02	0.08
	EMGNN	4.40	5.10	1.98	1.82	1.22	0.72	**2.62**	2.58	2.90	3.84	2.94	**1.18**
50	KAGNet	**8.76**	**7.32**	**4.30**	**3.44**	**1.78**	**0.96**	3.30	**4.22**	4.12	**5.18**	**4.74**	**1.36**
	EMOGI	3.06	1.34	2.04	0.76	0.44	0.20	0.92	1.08	1.36	1.54	1.40	0.34
	MTGCN	3.20	1.54	2.22	0.40	0.26	0.22	0.96	1.26	1.46	1.14	1.08	0.48
	DGHNN	4.52	3.28	3.64	1.46	0.48	0.72	1.12	3.82	3.28	0.94	2.86	0.84
	MODIG	3.96	5.80	3.46	1.00	0.00	0.08	1.46	1.72	3.26	3.32	1.70	0.12
	EMGNN	7.08	7.10	4.22	3.30	1.46	0.84	**3.74**	3.92	**5.00**	5.12	4.22	**1.36**

Table S5. Effect of knowledge-enriched gene representation on the fused KAGNet model across 12 cancer types. Results are reported as mean AUPRC ± standard deviation over 10 runs of stratified 5-fold cross-validation. The best result for each cancer type is shown in **bold**.

Cancer	Omics	HGT	Omics + HGT
BRCA	0.767 ± 0.007	0.790 ± 0.005	$\mathbf{0.835 \pm 0.004}$
BLCA	0.757 ± 0.010	0.628 ± 0.019	$\mathbf{0.798 \pm 0.008}$
LUAD	0.712 ± 0.007	0.585 ± 0.012	$\mathbf{0.742 \pm 0.006}$
LIHC	0.578 ± 0.015	0.505 ± 0.023	$\mathbf{0.619 \pm 0.017}$
THCA	0.375 ± 0.017	0.530 ± 0.017	$\mathbf{0.573 \pm 0.017}$
LUSC	0.506 ± 0.039	0.552 ± 0.045	$\mathbf{0.705 \pm 0.047}$
ESCA	0.570 ± 0.015	0.530 ± 0.014	$\mathbf{0.639 \pm 0.012}$
PRAD	0.784 ± 0.015	0.711 ± 0.020	$\mathbf{0.852 \pm 0.008}$
STAD	0.754 ± 0.007	0.535 ± 0.027	$\mathbf{0.755 \pm 0.011}$
COAD	0.608 ± 0.008	0.482 ± 0.009	$\mathbf{0.630 \pm 0.009}$
UCEC	0.741 ± 0.014	0.592 ± 0.011	$\mathbf{0.761 \pm 0.010}$
CESC	0.602 ± 0.037	0.626 ± 0.029	$\mathbf{0.717 \pm 0.036}$

Table S6. Effect of knowledge-enriched gene representation across individual PPI-specific branches. Results are reported as mean AUPRC over 10 runs of stratified 5-fold cross-validation. The best result for each cancer type and PPI network is shown in **bold**.

Cancer	Input	CPDB	STRING	MultiNet	PCNet	IRefIndex	IRefIndex2015
BRCA	Omics	0.713	0.639	0.654	0.714	0.632	0.628
	HGT	0.768	0.743	0.736	0.754	0.726	0.729
	Omics + HGT	**0.804**	**0.789**	**0.791**	**0.805**	**0.784**	**0.788**
BLCA	Omics	0.725	0.679	0.669	0.724	0.708	0.689
	HGT	0.546	**0.731**	0.478	0.517	0.543	0.465
	Omics + HGT	**0.749**	**0.731**	**0.699**	**0.780**	**0.745**	**0.718**
LUAD	Omics	0.662	0.648	0.610	0.671	0.629	0.607
	HGT	0.542	0.552	0.486	0.532	0.499	0.494
	Omics + HGT	**0.704**	**0.698**	**0.669**	**0.708**	**0.696**	**0.662**
LIHC	Omics	0.509	0.500	0.443	0.601	0.466	0.431
	HGT	0.473	0.466	0.414	0.449	0.430	0.408
	Omics + HGT	**0.576**	**0.544**	**0.518**	**0.615**	**0.551**	**0.513**
THCA	Omics	0.341	0.232	0.308	0.271	0.260	0.272
	HGT	0.478	0.452	0.419	0.458	0.441	0.439
	Omics + HGT	**0.506**	**0.478**	**0.465**	**0.504**	**0.453**	**0.478**
LUSC	Omics	0.394	0.381	0.348	0.429	0.368	0.348
	HGT	0.297	0.330	0.336	0.405	**0.513**	0.314
	Omics + HGT	**0.500**	**0.451**	**0.447**	**0.597**	0.511	**0.450**
ESCA	Omics	0.470	0.462	0.440	0.528	0.530	0.446
	HGT	0.424	0.460	0.402	0.467	0.490	0.435
	Omics + HGT	**0.572**	**0.593**	**0.542**	**0.600**	**0.619**	**0.563**
PRAD	Omics	0.705	0.679	0.581	0.693	0.694	0.652
	HGT	0.627	0.712	**0.709**	0.694	0.657	0.603
	Omics + HGT	**0.790**	**0.807**	**0.709**	**0.806**	**0.808**	**0.760**
STAD	Omics	0.681	**0.701**	0.667	0.722	0.703	0.666
	HGT	0.443	0.450	0.459	0.483	0.476	0.443
	Omics + HGT	**0.685**	0.665	**0.676**	**0.735**	**0.711**	**0.693**
COAD	Omics	0.559	0.540	0.532	0.602	0.541	0.542
	HGT	0.452	0.447	0.445	0.456	0.428	0.432
	Omics + HGT	**0.589**	**0.590**	**0.557**	**0.615**	**0.591**	**0.566**
UCEC	Omics	0.710	**0.700**	**0.666**	**0.727**	0.689	0.650
	HGT	0.554	0.462	0.482	0.525	0.532	0.510
	Omics + HGT	**0.725**	0.689	0.663	0.713	**0.713**	**0.697**
CESC	Omics	0.476	0.385	0.358	0.418	0.445	0.365
	HGT	0.442	**0.569**	0.418	0.346	0.515	0.331
	Omics + HGT	**0.555**	**0.569**	**0.434**	**0.592**	**0.550**	**0.452**

Table S7. Effect of AlphaGenome-guided pretraining on the fused KAGNet model across 12 cancer types. Results are reported as mean AUPRC ± standard deviation over 10 runs of stratified 5-fold cross-validation. The best result for each cancer type is shown in **bold**.

Cancer	Without AlphaGenome	With AlphaGenome
BRCA	0.833 ± 0.005	$\mathbf{0.835 \pm 0.004}$
BLCA	$\mathbf{0.799 \pm 0.008}$	0.798 ± 0.008
LUAD	$\mathbf{0.742 \pm 0.008}$	$\mathbf{0.742 \pm 0.006}$
LIHC	0.608 ± 0.017	$\mathbf{0.619 \pm 0.017}$
THCA	0.569 ± 0.022	$\mathbf{0.573 \pm 0.017}$
LUSC	0.582 ± 0.046	$\mathbf{0.705 \pm 0.047}$
ESCA	0.628 ± 0.011	$\mathbf{0.639 \pm 0.012}$
PRAD	$\mathbf{0.852 \pm 0.008}$	$\mathbf{0.852 \pm 0.008}$
STAD	0.745 ± 0.013	$\mathbf{0.755 \pm 0.011}$
COAD	$\mathbf{0.632 \pm 0.008}$	0.630 ± 0.009
UCEC	0.752 ± 0.014	$\mathbf{0.761 \pm 0.010}$
CESC	0.604 ± 0.031	$\mathbf{0.717 \pm 0.036}$

Table S8. Effect of AlphaGenome-guided pretraining across individual PPI-specific branches. Results are reported as mean AUPRC over 10 runs of stratified 5-fold cross-validation. The better result within each PPI network is shown in **bold**.

Cancer	CPDB		STRING		MultiNet		PCNet		IRefIndex		IRefIndex2015	
	No AG	With AG	No AG	With AG	No AG	With AG	No AG	With AG	No AG	With AG	No AG	With AG
BRCA	0.804	**0.820**	0.789	**0.790**	**0.791**	0.782	0.805	**0.819**	0.784	**0.786**	**0.788**	0.779
BLCA	0.749	**0.754**	0.731	**0.732**	**0.699**	0.695	**0.780**	0.770	**0.745**	0.738	**0.718**	0.704
LUAD	**0.704**	0.701	**0.698**	0.696	**0.669**	0.664	**0.708**	0.705	**0.696**	0.690	0.662	**0.665**
LIHC	0.576	**0.592**	0.544	**0.545**	**0.518**	0.514	0.615	**0.618**	**0.551**	0.540	0.513	**0.518**
THCA	0.506	**0.511**	**0.478**	0.476	0.465	**0.473**	0.504	**0.511**	**0.453**	0.445	**0.478**	0.465
LUSC	0.500	**0.550**	0.451	**0.502**	0.447	**0.540**	0.597	**0.654**	0.511	**0.632**	0.450	**0.482**
ESCA	0.572	**0.589**	0.593	**0.595**	0.542	**0.545**	0.600	**0.604**	**0.619**	0.615	0.563	**0.572**
PRAD	0.790	**0.798**	**0.807**	0.803	0.709	**0.742**	**0.806**	0.800	0.808	**0.814**	0.760	**0.779**
STAD	0.685	**0.695**	0.665	**0.667**	0.676	**0.678**	**0.735**	0.732	0.711	**0.714**	**0.693**	0.689
COAD	0.589	**0.594**	**0.590**	0.586	**0.557**	0.553	**0.615**	0.611	0.591	**0.595**	**0.566**	0.556
UCEC	**0.725**	0.715	0.689	**0.694**	0.663	**0.678**	0.713	**0.726**	0.713	**0.718**	**0.697**	0.685
CESC	0.555	**0.694**	0.569	**0.629**	0.434	**0.500**	0.592	**0.668**	0.550	**0.669**	0.452	**0.509**

Table S9. AUPRC comparison between KAGNet and KAGNet-Rand across 12 cancer types. Results are reported as mean AUPRC $\pm$ standard deviation over 10 runs of stratified 5-fold cross-validation. KAGNet-Rand denotes the control variant in which the AlphaGenome vectors used for encoder pretraining were replaced by random vectors drawn independently for each gene from a Gaussian distribution whose per-dimension mean and standard deviation match those of the original AlphaGenome feature matrix. The random matrix was generated once (fixed seed) and used to pretrain the encoders of all PPI networks; all subsequent steps were identical to KAGNet. The higher mean for each cancer type is shown in **bold** (both are bolded in case of a tie).

Cancer	KAGNet	KAGNet-Rand
BRCA	0.835 ± 0.004	$\mathbf{0.837 \pm 0.003}$
BLCA	$\mathbf{0.798 \pm 0.008}$	0.797 ± 0.007
LUAD	0.742 ± 0.006	$\mathbf{0.744 \pm 0.006}$
LIHC	$\mathbf{0.619 \pm 0.017}$	0.618 ± 0.018
THCA	$\mathbf{0.573 \pm 0.017}$	0.569 ± 0.022
LUSC	$\mathbf{0.705 \pm 0.047}$	0.673 ± 0.042
ESCA	0.639 ± 0.012	$\mathbf{0.640 \pm 0.012}$
PRAD	0.852 ± 0.008	$\mathbf{0.860 \pm 0.010}$
STAD	$\mathbf{0.755 \pm 0.011}$	0.749 ± 0.007
COAD	0.630 ± 0.009	$\mathbf{0.632 \pm 0.009}$
UCEC	0.761 ± 0.010	$\mathbf{0.762 \pm 0.010}$
CESC	$\mathbf{0.717 \pm 0.036}$	0.699 ± 0.034

Table S10. Comparison of Hits@K between KAGNet and KAGNet-Rand for LUSC and CESC at $K = 1, 3, 5, 10, 20, 50$. KAGNet-Rand denotes the control variant in which the AlphaGenome vectors used for encoder pretraining were replaced by random vectors drawn independently for each gene from a Gaussian distribution whose per-dimension mean and standard deviation match those of the original AlphaGenome feature matrix. The random matrix was generated once (fixed seed) and used to pretrain the encoders of all PPI networks; all subsequent steps were identical to KAGNet. Hits@K for KAGNet-Rand was computed for the two cancer types with the largest pretraining gains. The better result of the two variants for each cancer type at each K is shown in **bold** (both are bolded in case of a tie).

K	Method	LUSC	CESC
1	KAGNet	**0.18**	0.24
	KAGNet-Rand	0.16	**0.30**
3	KAGNet	0.20	**0.46**
	KAGNet-Rand	**0.26**	**0.46**
5	KAGNet	**0.36**	**0.64**
	KAGNet-Rand	**0.36**	0.50
10	KAGNet	0.58	**0.80**
	KAGNet-Rand	**0.60**	0.72
20	KAGNet	**0.76**	1.08
	KAGNet-Rand	0.74	**1.10**
50	KAGNet	**0.96**	**1.36**
	KAGNet-Rand	0.94	1.32

Table S11. Paired comparison of KAGNet and EMGNN trained on the same omics and HGT features (z-score normalized). Values are mean AUPRC ± standard deviation over 10 runs. Runs are paired by seed, so that both methods use the same per-run seed for the cross-validation split. Δ is the mean of the run-wise differences (KAGNet − EMGNN) with its 95% CI from the paired t distribution; r is the Pearson correlation between the two methods' run-level AUPRC; p is the two-sided paired t-test p-value, p_{Holm} is adjusted across the 12 cancers, and bold values with * indicate $p_{\text{Holm}} < 0.05$. Across cancers (paired by cancer), KAGNet is higher in 7 of 12 cancers (median $\Delta = +0.008$; paired t $p = 0.941$; Wilcoxon $p = 0.569$), and in 7 of 11 when CESC is excluded (median $\Delta = +0.009$; paired t $p = 0.188$; Wilcoxon $p = 0.240$).

Cancer	KAGNet	EMGNN+HGT	Δ [95% CI]	r	p	p_{Holm}
BRCA	0.835 ± 0.004	0.825 ± 0.006	$+\mathbf{0.010}^{*}$ [+0.006, +0.015]	+0.45	2.9×10^{-4}	**0.003**
BLCA	0.798 ± 0.008	0.789 ± 0.008	+0.009 [−0.001, +0.020]	−0.50	0.079	0.334
LUAD	0.742 ± 0.006	0.727 ± 0.015	+0.015 [+0.005, +0.025]	+0.48	0.009	0.080
LIHC	0.619 ± 0.017	0.581 ± 0.029	+0.038 [+0.013, +0.063]	+0.09	0.007	0.068
THCA	0.573 ± 0.017	0.553 ± 0.025	+0.020 [+0.006, +0.034]	+0.68	0.011	0.085
LUSC	0.705 ± 0.047	0.663 ± 0.057	+0.043 [−0.004, +0.090]	+0.30	0.067	0.334
ESCA	0.639 ± 0.012	0.665 ± 0.022	$-\mathbf{0.026}^{*}$ [−0.039, −0.013]	+0.64	0.001	**0.014**
PRAD	0.852 ± 0.008	0.869 ± 0.018	−0.017 [−0.031, −0.003]	+0.27	0.020	0.141
STAD	0.755 ± 0.011	0.756 ± 0.011	0.000 [−0.013, +0.012]	−0.15	0.945	1.000
COAD	0.630 ± 0.009	0.631 ± 0.012	−0.002 [−0.015, +0.012]	−0.43	0.789	1.000
UCEC	0.761 ± 0.010	0.754 ± 0.012	+0.007 [−0.002, +0.016]	+0.46	0.105	0.334
CESC	0.717 ± 0.036	0.805 ± 0.082	−0.088 [−0.159, −0.017]	−0.15	0.021	0.141

Table S12. Comparison of Hits@K between EMGNN trained on omics and HGT features (z-score normalized) and KAGNet across 12 cancer types for $K = 1, 3, 5, 10, 20, 50$. Each value is the mean number of known driver genes recovered among the top-K ranked candidates per fold, averaged over 10 runs × 5 folds. The higher value for each cancer type at each K is shown in **bold**.

K	Method	BRCA	BLCA	LUAD	LIHC	THCA	LUSC	ESCA	PRAD	STAD	COAD	UCEC	CESC
1	KAGNet	**0.66**	0.28	0.42	**0.28**	**0.48**	**0.18**	0.78	**0.40**	0.48	**0.92**	**0.78**	**0.24**
	EMGNN+HGT	0.52	**0.42**	**0.44**	0.24	0.40	0.16	**0.80**	0.16	**0.54**	0.82	0.76	0.20
3	KAGNet	**1.48**	**1.66**	0.70	**0.86**	**0.76**	0.20	**1.42**	**1.22**	**1.16**	**2.16**	**1.34**	0.46
	EMGNN+HGT	1.26	1.54	**0.78**	0.70	0.62	**0.28**	**1.42**	1.08	1.06	1.74	1.24	**0.48**
5	KAGNet	**2.30**	**2.62**	**1.06**	**1.00**	**1.02**	0.36	**1.70**	**1.54**	**1.36**	**2.60**	1.56	**0.64**
	EMGNN+HGT	1.92	2.36	1.04	0.88	0.70	**0.38**	1.60	1.22	1.32	2.20	**1.62**	0.62
10	KAGNet	**4.14**	**4.26**	**1.62**	**1.30**	**1.10**	0.58	**2.04**	**1.98**	1.58	**3.38**	2.30	0.80
	EMGNN+HGT	3.40	3.68	1.40	1.14	1.08	**0.60**	1.96	1.78	**1.84**	2.84	**2.50**	**1.02**
20	KAGNet	5.42	**5.68**	**2.06**	2.02	1.28	**0.76**	2.60	2.84	**3.04**	**4.28**	**3.32**	1.08
	EMGNN+HGT	**5.58**	5.30	1.90	**2.10**	**1.34**	0.72	**2.66**	**3.16**	2.92	3.92	3.16	**1.14**
50	KAGNet	8.76	7.32	**4.30**	**3.44**	1.78	0.96	3.30	4.22	4.12	5.18	4.74	1.36
	EMGNN+HGT	**9.06**	**7.40**	4.24	3.02	**1.90**	**1.08**	**3.60**	**4.30**	**4.32**	**5.22**	**5.04**	**1.52**

Table S13. Exploratory paired comparison of Hits@K between KAGNet and EMGNN trained on the same omics and HGT features. Each entry is Δ, the mean Hits@K per fold of KAGNet minus that of EMGNN, averaged over 10 runs; positive values favor KAGNet. Runs are paired by seed, so that both methods use the same per-run seed for the cross-validation split. Bold values with * indicate a Holm-adjusted $p < 0.05$ (two-sided paired t-test on run-level means, adjusted across the 12 cancers within each K). The lower block summarizes each K across cancers, for all 12 cancers and with CESC excluded: the number of cancers in which KAGNet is higher, the median paired effect size d_z (mean of the run-wise differences divided by their standard deviation), the Wilcoxon signed-rank p-value on d_z across cancers, and this p-value Holm-adjusted over the six values of K.

Cancer	K = 1	K = 3	K = 5	K = 10	K = 20	K = 50
BRCA	+0.14	+0.22	**+0.38***	**+0.74***	−0.16	−0.30
BLCA	−0.14	+0.12	+0.26	**+0.58***	+0.38	−0.08
LUAD	−0.02	−0.08	+0.02	+0.22	+0.16	+0.06
LIHC	+0.04	+0.16	+0.12	+0.16	−0.08	+0.42
THCA	+0.08	+0.14	**+0.32***	+0.02	−0.06	−0.12
LUSC	+0.02	−0.08	−0.02	−0.02	+0.04	−0.12
ESCA	−0.02	0.00	+0.10	+0.08	−0.06	−0.30
PRAD	**+0.24***	+0.14	**+0.32***	**+0.20***	−0.32	−0.08
STAD	−0.06	+0.10	+0.04	**−0.26***	+0.12	−0.20
COAD	+0.10	**+0.42***	+0.40	**+0.54***	+0.36	−0.04
UCEC	+0.02	+0.10	−0.06	−0.20	+0.16	−0.30
CESC	+0.04	−0.02	+0.02	**−0.22***	−0.06	−0.16
All 12 cancers						
KAGNet higher	8/12	8/12	10/12	8/12	6/12	2/12
Median d_z	+0.32	+0.72	+0.50	+0.62	+0.06	−0.35
Wilcoxon p	0.176	0.052	0.012	0.301	0.258	0.027
p_{Holm} over K	0.529	0.209	0.073	0.529	0.529	0.134
Excluding CESC						
KAGNet higher	7/11	8/11	9/11	8/11	6/11	2/11
p_{Holm} over K	0.442	0.210	0.082	0.442	0.442	0.215

Table S14. Per-cancer details of the paired Hits@K comparison between KAGNet and EMGNN trained on the same omics and HGT features. Each cell reports P_{win} / p_{Holm} for a given cancer and K. P_{win} is the proportion of the 10 seed-matched run pairs in which KAGNet has a higher mean Hits@K than EMGNN, with ties counted as one half (0.5 indicates no difference). p_{Holm} is the two-sided paired t-test p-value on run-level means, adjusted across the 12 cancers within each K; bold cells indicate $p_{\text{Holm}} < 0.05$.

Cancer	K = 1	K = 3	K = 5	K = 10	K = 20	K = 50
BRCA	0.70/0.775	0.75/0.294	**0.90/0.025**	**0.90/0.008**	0.45/1.000	0.30/1.000
BLCA	0.20/0.730	0.80/0.056	0.85/0.085	**0.95/0.008**	0.85/0.053	0.45/1.000
LUAD	0.40/1.000	0.35/0.669	0.50/1.000	0.75/0.191	0.70/1.000	0.55/1.000
LIHC	0.60/1.000	0.75/0.224	0.75/0.568	0.75/0.191	0.45/1.000	0.80/0.140
THCA	0.70/0.405	0.80/0.224	**0.95/0.006**	0.55/1.000	0.40/1.000	0.30/0.649
LUSC	0.55/1.000	0.30/0.294	0.50/1.000	0.45/1.000	0.60/1.000	0.35/1.000
ESCA	0.45/1.000	0.50/1.000	0.65/1.000	0.65/0.504	0.50/1.000	0.35/0.670
PRAD	**0.90/0.012**	0.75/0.294	**0.95/0.012**	**0.85/0.026**	0.25/0.224	0.50/1.000
STAD	0.35/0.730	0.65/0.553	0.60/1.000	**0.10/0.008**	0.65/1.000	0.15/0.102
COAD	0.75/0.522	**0.95/0.029**	0.80/0.105	**0.90/0.013**	0.85/0.110	0.40/1.000
UCEC	0.55/1.000	0.75/0.294	0.35/1.000	0.25/0.469	0.75/0.224	0.10/0.307
CESC	0.60/1.000	0.45/1.000	0.50/1.000	**0.10/0.026**	0.35/1.000	0.20/0.118